

John Liu

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EDUCATION & AWARDS

University of Waterloo

Computer Science, 3.9/4.0 Major GPA

Waterloo, ON

Expected April 2028

- Canadian Mathematical Olympiad Qualifier, AIME Qualifier (5×)

WORK EXPERIENCE

Jane Street

Incoming Software Engineer Intern

New York, NY

May 2027 – Aug 2027

Figma

Software Engineer Intern

San Francisco, CA

May 2026 – Aug 2026

- Shipped composed color and opacity expressions across a **TypeScript** and **React** editor and **C++/WebAssembly** runtime, adding extensible infrastructure for computed relationships between design variables
- Designed **expression semantics** and context-aware variable resolution within a centralized JSON serialization layer, enabling expression support across customer facing **plugin** and **REST APIs**
- Reduced variable mutation propagation latency by **29%** by eliminating redundant dependency graph traversals through an **event-driven dependency system** and caching expression alias resolutions

WindBorne Systems Inc.

Software Engineer Intern

Palo Alto, CA

May 2025 – Aug 2025

- Owned development for a meteorological insights platform used by energy traders, building features across **TypeScript**, **Rails**, **Rust**, and **WebGL** systems supporting a product generating **\$2.5M ARR**
- Implemented tropical cyclone forecasting algorithms in **C++** and **Python** over **200TB+** of monthly global weather data, reducing query latency by **28%** through a custom geospatial caching layer
- Built performant weather visualization features for a **Rust/WebGL** graphing engine, maintaining interactive frame rates while rendering millions of forecast data points in the browser

CIBC

Software Engineer Intern

Toronto, ON

Jan 2025 – Apr 2025

- Delivered real-time chatbot features using **Python**, **Redis**, and websocket streaming, enabling low-latency tool execution and interactive user experiences for CIBC's internal AI assistant
- Engineered LLM-powered capabilities including **RAG** over patent databases, live news ingestion pipelines, and **sentiment classification** systems for document triage and automated summarization
- Architected a full-stack LLM prototyping framework (**Next.js**, **Prisma**, **FastAPI**) adopted across **10+** internal repositories, standardizing deployment and evaluation workflows

PROJECTS

unfollowed.lol | JavaScript, PostgreSQL, Django

- Launched a Chrome extension for tracking marketing analytics that reached **#8** on Product Hunt with **100+** upvotes and reached **4,000+** daily active users

SlimeScholars | React, Next.js, MongoDB, AWS

- Built an educational content platform with a custom lesson builder supporting **LaTeX**, **XML**, and custom markdown-based authoring workflows, with analytics dashboards tracking student engagement across lessons

SKILLS

Languages: Python, C++, TypeScript, Rust, Go, Java, Ruby, SQL

Frameworks: React, Next.js, Rails, FastAPI, PostgreSQL, Docker, Kubernetes, Redis, MongoDB, AWS

Areas: Operating Systems, Distributed Systems, WebGL, Algorithms & Data Structures, Networking, Concurrency